

Global AI & Data Science Job Market Analysis 2026

An End-to-End Business Intelligence Project Using Python, SQL & Power BI

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Tools & Technologies

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - SQL (MySQL)
 - Power BI
 - Microsoft Excel
 - Git & GitHub
-

Submitted As

Professional Data Analytics Portfolio Project

Project Domain

Artificial Intelligence • Data Science • Human Resources Analytics • Business Intelligence

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Acknowledgement

This project represents the successful completion of an end-to-end Business Intelligence and Data Analytics workflow. It demonstrates practical application of Python, SQL, and Power BI to solve a real-world business problem involving salary benchmarking within the global Artificial Intelligence and Data Science job market.

I would like to acknowledge the Kaggle community for providing the dataset that made this analysis possible. I also extend my gratitude to the open-source Python ecosystem—including Pandas, NumPy, Matplotlib, and Seaborn—as well as Microsoft Power BI and MySQL for providing the analytical tools used throughout this project.

This report reflects my commitment to continuous learning, analytical thinking, and transforming raw data into actionable business insights.

Executive Summary

The rapid expansion of Artificial Intelligence (AI) and Data Science has significantly transformed the global employment landscape, creating unprecedented demand for highly skilled professionals. As organizations increasingly invest in AI-driven technologies, understanding salary trends, workforce characteristics, and compensation drivers has become essential for business leaders, HR professionals, recruiters, and job seekers.

This project presents a comprehensive end-to-end analysis of the **AI & Data Science Job Salaries 2026** dataset using Python, SQL, and Microsoft Power BI. The primary objective is to identify the factors influencing salary levels across different job roles, industries, company sizes, experience levels, and geographic regions. By integrating data preparation, exploratory data analysis, business intelligence techniques, and interactive visualization, the project provides meaningful insights that support strategic workforce planning and evidence-based decision-making.

The analytical workflow begins with data acquisition and preprocessing using Python. Data quality was assessed through missing value analysis, duplicate detection, and validation of key variables. Feature engineering techniques were then applied to create new analytical variables, enabling deeper segmentation of salary ranges, experience groups, certification levels, and AI tool usage patterns.

Following data preparation, an extensive Exploratory Data Analysis (EDA) was conducted to uncover hidden trends, identify salary distributions, examine industry-specific compensation, evaluate remote work adoption, and assess the relationship between experience and earnings. Statistical techniques were employed to summarize key characteristics of the dataset and strengthen the reliability of the findings.

Business-oriented SQL analysis was performed to answer practical questions regarding salary benchmarking, top-paying job roles, geographic salary differences, company size comparisons, and workforce trends. Advanced SQL concepts, including aggregate functions, grouping, ranking, and window functions, were used to extract meaningful business intelligence from the dataset.

To enhance decision-making capabilities, an executive-level Power BI dashboard was developed. The dashboard includes interactive KPI cards, salary analysis, geographic visualization, company size comparisons, industry insights, and dynamic filtering options, allowing stakeholders to explore the data efficiently.

The findings indicate that experience level, organizational size, industry sector, geographic location, and AI specialization significantly influence compensation. Senior and executive professionals consistently receive higher salaries, while organizations investing heavily in AI technologies tend to offer more competitive compensation packages. The analysis also highlights the increasing importance of certifications, AI tool proficiency, and flexible work arrangements in shaping career opportunities.

Overall, this project demonstrates an end-to-end Business Intelligence workflow that combines technical analysis with business storytelling. Beyond showcasing proficiency in Python, SQL, and Power BI, the report emphasizes the ability to transform raw data into actionable insights, making it suitable for professional portfolios and international Data Analyst, Business Intelligence Analyst, and Junior Data Scientist roles.

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2. Business Problem

2.1 Background

Artificial Intelligence (AI) and Data Science have become fundamental drivers of digital transformation across industries. Organizations worldwide are investing heavily in AI-powered solutions to improve operational efficiency, automate decision-making, and gain a competitive advantage. As demand for AI professionals continues to rise, compensation structures have become increasingly complex, varying significantly across job roles, industries, company sizes, geographic regions, and experience levels.

For employers, designing competitive compensation packages is essential to attract and retain top talent. For professionals, understanding market salary trends helps in making informed career decisions. However, without a comprehensive data-driven analysis, salary benchmarking often relies on fragmented information, making strategic workforce planning challenging.

This project addresses these challenges by analyzing the **AI & Data Science Job Salaries 2026** dataset to identify the key factors influencing compensation. Through Python, SQL, and Power BI, the project transforms raw salary records into actionable business intelligence that supports evidence-based decision-making.

2.2 Problem Statement

Organizations, HR professionals, and job seekers often struggle to understand the rapidly evolving AI job market due to variations in salaries across experience levels, industries, company sizes, geographic locations, and remote work arrangements. The absence of centralized analytical insights makes it difficult to benchmark compensation, identify high-paying opportunities, and formulate effective talent acquisition strategies.

This project seeks to bridge this gap by conducting an end-to-end analytical assessment of AI and Data Science salary data. The analysis aims to uncover compensation patterns, workforce trends, and the impact of professional qualifications on earnings while providing meaningful recommendations for strategic decision-making.

2.3 Business Questions

The analysis focuses on answering the following key business questions:

- What is the average salary of AI and Data Science professionals?
 - Which job roles offer the highest compensation?
 - How does salary vary across experience levels?
 - Does company size influence employee compensation?
 - Which countries provide the highest-paying opportunities?
 - Which industries offer the most competitive salaries?
 - Does remote work affect salary levels?
 - How do certifications influence compensation?
 - Is frequent AI tool usage associated with higher salaries?
 - What factors contribute most significantly to employee earnings?
-

3. Project Objectives

3.1 Primary Objective

The primary objective of this project is to perform an end-to-end business intelligence analysis of the AI and Data Science job market by leveraging Python, SQL, and Microsoft Power BI. The project aims to transform raw employment data into meaningful business insights that support salary benchmarking, workforce planning, and strategic decision-making.

3.2 Specific Objectives

The project was designed to achieve the following objectives:

- Assess the overall quality of the dataset through comprehensive data profiling.
 - Identify and resolve missing values, duplicate records, and inconsistent data.
 - Perform feature engineering to enhance analytical capabilities.
 - Explore salary distributions across job roles, industries, company sizes, and geographic regions.
 - Examine the relationship between experience level and compensation.
 - Analyze the impact of certifications, AI tool usage, and remote work on salary outcomes.
 - Conduct business-oriented SQL analysis using aggregate functions, grouping, ranking, and window functions.
 - Develop an interactive executive dashboard in Microsoft Power BI.
 - Generate actionable business insights and strategic recommendations.
 - Demonstrate an end-to-end analytics workflow suitable for professional portfolio presentation.
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3.3 Scope of the Project

The scope of this project includes descriptive analytics, business intelligence reporting, and dashboard development based on historical AI and Data Science employment data. The analysis focuses exclusively on understanding salary patterns and workforce characteristics and does not involve predictive modeling or machine learning.

The project covers the complete analytics lifecycle, including data preparation, exploratory data analysis, SQL-based business analysis, dashboard development, and strategic reporting.

4. Dataset Overview

4.1 Dataset Description

The dataset used in this project is **AI & Data Science Job Salaries 2026**, sourced from Kaggle. It contains detailed information on employment, compensation, experience, organizational characteristics, and workforce demographics related to AI and Data Science professionals.

The dataset serves as a valuable resource for analyzing salary trends and understanding the factors that influence compensation in one of the fastest-growing technology sectors.

4.2 Dataset Source

Source: Kaggle

Dataset Name: AI & Data Science Job Salaries 2026

File Format: CSV

Domain: Human Resources Analytics / Business Intelligence

4.3 Dataset Characteristics

Attribute	Description
Dataset Type	Structured Tabular Data
File Format	CSV
Domain	Artificial Intelligence & Human Resources
Number of Records	Approximately 5,000
Number of Variables	27
Analysis Type	Descriptive & Business Intelligence

4.4 Key Variables

The dataset contains information related to:

- Job Title
- Experience Level
- Employment Type
- Company Size
- Company Location
- Employee Residence
- Industry
- Remote Work Ratio
- Years of Experience
- Education Level
- Primary Programming Language
- AI Tool Usage
- Certifications

- Weekly Working Hours
- Salary (USD)
- Bonus Percentage
- Equity Offered
- Job Satisfaction Score
- Team Size
- Interview-to-Offer Ratio
- Career Mobility Indicators

These variables enable multidimensional analysis of compensation trends and workforce dynamics.

4.5 Business Relevance

The dataset provides valuable insights for multiple stakeholders:

Human Resource Departments

- Benchmark salary structures
- Improve compensation planning
- Optimize talent acquisition strategies

Business Leaders

- Understand labor market dynamics
- Support workforce planning
- Identify competitive salary ranges

Job Seekers

- Evaluate market compensation
- Identify high-paying industries and roles
- Plan career progression

Data Analysts

- Practice end-to-end analytics workflows
 - Develop dashboarding and reporting skills
 - Demonstrate business intelligence capabilities
-

4.6 Expected Project Outcomes

Upon completion, the project is expected to:

- Identify the major factors influencing AI salaries.
- Reveal salary differences across industries, countries, and company sizes.
- Highlight high-paying job roles and emerging market trends.
- Provide an executive dashboard for interactive analysis.
- Deliver actionable recommendations to support data-driven decision-making.
- Showcase practical expertise in Python, SQL, and Power BI through a professional portfolio project.

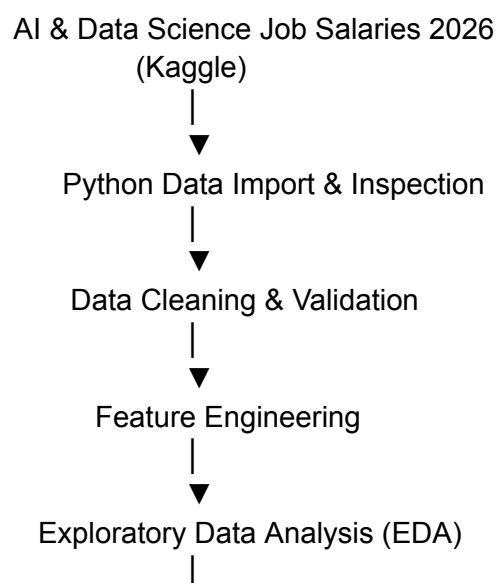
5. Project Architecture

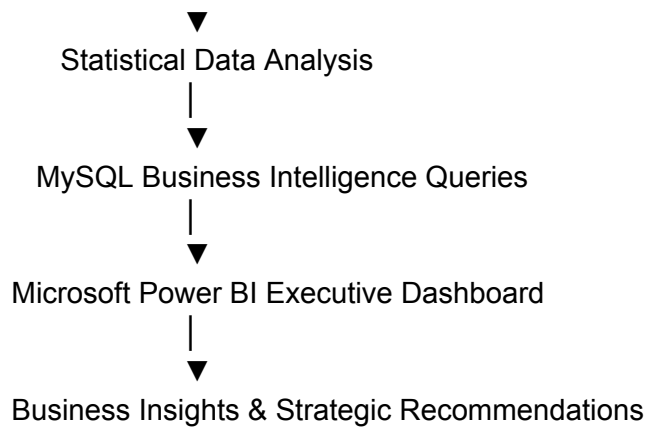
5.1 Project Overview

This project follows a structured end-to-end Business Intelligence workflow that transforms raw employment data into meaningful business insights. Rather than focusing solely on visualization, the project demonstrates the complete analytics lifecycle, beginning with data acquisition and ending with executive-level reporting.

The architecture was designed to ensure data quality, analytical accuracy, and reproducibility while showcasing practical proficiency in Python, SQL, and Microsoft Power BI. Each stage builds upon the previous one, resulting in a scalable analytics pipeline suitable for professional portfolio presentation.

5.2 Analytics Workflow





5.3 Technology Stack

The project integrates multiple analytical technologies to perform different stages of the data analytics lifecycle.

Technology	Purpose
Python	Data Cleaning, Feature Engineering, EDA
Pandas	Data Manipulation
NumPy	Numerical Computation
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
MySQL	Business Analysis
Power BI	Interactive Dashboard
Git & GitHub	Version Control & Portfolio
VS Code	Development Environment

6. Python Data Preparation & Feature Engineering

6.1 Introduction

Data preparation is one of the most critical stages of any analytics project. High-quality insights depend on the reliability and consistency of the underlying data. Therefore, before conducting any business analysis, the dataset was carefully inspected, validated, and transformed using Python.

This stage involved importing the dataset, assessing its structure, validating data quality, identifying inconsistencies, and creating new analytical features to improve business understanding.

6.2 Data Import

The dataset was imported into Python using the Pandas library, which provides efficient tools for handling structured tabular data.

The initial inspection included:

- Loading the CSV dataset
- Displaying sample records
- Checking dataset dimensions
- Identifying data types
- Reviewing column names
- Understanding overall dataset structure

This initial assessment established the foundation for subsequent data cleaning and analysis.

6.3 Data Inspection

A comprehensive data inspection was performed to evaluate the overall quality and completeness of the dataset.

The following aspects were examined:

- Number of observations
- Number of variables
- Missing values
- Duplicate records
- Numerical summaries
- Categorical distributions
- Memory usage
- Data consistency

This inspection ensured that the dataset was suitable for further analytical processing.

6.4 Data Quality Assessment

Maintaining high data quality is essential for producing accurate business insights. A structured assessment was conducted to evaluate the integrity of the dataset.

The assessment focused on the following dimensions:

Completeness

All records were examined to identify missing or incomplete values.

Consistency

Categorical variables were reviewed for inconsistent naming conventions and formatting.

Accuracy

Salary values and numerical attributes were validated to ensure they fell within logical business ranges.

Uniqueness

Duplicate records were identified and evaluated to avoid biased analytical results.

Validity

Each feature was checked against its expected data type and business meaning.

6.5 Missing Value Analysis

Missing values can significantly influence analytical outcomes if not handled appropriately. A systematic review of the dataset was conducted to identify incomplete observations.

The analysis involved:

- Detecting missing values across all columns.
- Measuring the percentage of missing data.
- Assessing the business impact of incomplete records.
- Determining appropriate treatment strategies.

Depending on the characteristics of each variable, missing values were either retained, imputed, or excluded to preserve analytical reliability.

6.6 Duplicate Record Analysis

Duplicate observations can distort statistical summaries and business conclusions.

To ensure data integrity:

- Duplicate records were identified.
- Duplicate frequency was measured.
- Potential causes were investigated.
- Redundant records were removed where appropriate.

This process ensured that every observation represented a unique employment record.

6.7 Data Type Validation

Correct data types are essential for accurate calculations and visualizations.

The following validations were performed:

- Numeric variables
- Categorical variables
- Boolean variables
- Percentage fields
- Salary fields

Necessary conversions were applied to ensure consistency throughout the analytical workflow.

6.8 Feature Engineering

Feature engineering enhances the analytical value of the dataset by creating new variables that provide additional business context.

Several derived features were introduced to simplify segmentation and improve interpretability.

Salary Category

Employees were grouped into salary bands to facilitate compensation analysis.

Example:

- Low Salary
- Medium Salary

- High Salary
 - Executive Salary
-

Experience Group

Years of experience were transformed into broader career stages.

Example:

- Entry Level
 - Junior Professional
 - Mid-Level Professional
 - Senior Professional
 - Executive Leadership
-

AI Usage Level

AI tool usage was categorized to examine the relationship between technology adoption and employee compensation.

Example:

- Low Usage
 - Moderate Usage
 - High Usage
-

Certification Level

Professional certifications were grouped into meaningful learning categories to evaluate their influence on salary.

Example:

- No Certification
 - Beginner
 - Intermediate
 - Advanced
-

Workload Category

Weekly working hours were segmented to analyze employee workload patterns.

Example:

- Part-Time
 - Standard
 - Extended Hours
 - Intensive Workload
-

6.9 Data Validation After Feature Engineering

Following feature creation, the transformed dataset underwent an additional validation process to ensure analytical accuracy.

The validation included:

- Reviewing newly created variables.
 - Checking category distributions.
 - Confirming logical consistency.
 - Verifying numerical calculations.
 - Ensuring compatibility with Power BI and SQL.
-

6.10 Exporting the Final Dataset

After completing data preparation and feature engineering, the cleaned dataset was exported for downstream analysis.

The processed dataset served as the primary input for:

- Exploratory Data Analysis (EDA)
- Statistical Analysis
- SQL Business Intelligence
- Power BI Dashboard Development

This ensured that all subsequent analyses were performed using a consistent and validated data source.

6.11 Deliverables of This Phase

The Python data preparation stage produced the following outputs:

Deliverable	Purpose
-------------	---------

Cleaned Dataset	Reliable input for analysis
Feature-Engineered Dataset	Enhanced analytical capability
Validation Report	Data quality assurance
Python Scripts	Reproducible workflow
Summary Statistics	Initial business understanding

6.12 Chapter Summary

The Python data preparation stage established a strong analytical foundation by transforming raw employment data into a clean, consistent, and business-ready dataset. Through systematic validation, feature engineering, and quality assessment, the dataset became suitable for advanced statistical analysis, SQL-based business intelligence, and interactive dashboard development. This phase highlights the importance of robust data preparation in ensuring that subsequent analytical findings are accurate, reliable, and actionable.

7. Exploratory Data Analysis

7.1 Introduction

Exploratory Data Analysis (EDA) is a fundamental stage in the data analytics lifecycle that focuses on understanding the underlying structure, patterns, relationships, and anomalies within a dataset before conducting advanced analysis. Rather than relying on assumptions, EDA provides evidence-based insights through descriptive statistics and visual exploration.

For this project, EDA was performed using Python libraries including **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn**. The analysis aimed to examine salary distributions, workforce characteristics, company demographics, geographic trends, and employment patterns within the global Artificial Intelligence and Data Science job market.

The findings from this chapter establish the analytical foundation for subsequent SQL business analysis and Power BI dashboard development.

7.2 Objectives of Exploratory Data Analysis

The primary objectives of the EDA phase were to:

- Understand the distribution of salaries across the AI job market.
 - Identify relationships between experience level and compensation.
 - Compare salaries across company sizes and industries.
 - Analyze geographic variations in employee compensation.
 - Evaluate remote work adoption.
 - Examine the influence of certifications and AI tool usage.
 - Detect outliers and unusual observations.
 - Identify hidden business trends that support strategic decision-making.
-

7.3 Salary Distribution Analysis

Objective

To understand how salaries are distributed across the AI and Data Science workforce and identify the presence of skewness, concentration, or extreme values.

Visualization

Histogram – Salary Distribution

(Insert your Python histogram here.)

Interpretation

The salary distribution indicates a positively skewed pattern, with the majority of professionals earning within the mid-range salary bracket while a relatively small proportion receive exceptionally high compensation. This suggests that executive-level and highly specialized AI positions command substantial salary premiums compared to the broader workforce.

Business Insight

The presence of a right-skewed salary distribution highlights the significant earning potential associated with advanced AI expertise and leadership roles. Organizations should recognize these market dynamics when designing compensation strategies for high-demand talent.

7.4 Experience Level Analysis

Objective

To evaluate the relationship between professional experience and annual salary.

Visualization

Bar Chart – Average Salary by Experience Level

(Insert your Python chart here.)

Interpretation

A clear positive relationship exists between years of experience and average salary. Entry-level professionals receive the lowest compensation, while executive and leadership positions consistently achieve the highest salary levels.

Business Insight

Experience remains one of the strongest determinants of compensation. Organizations should implement structured career progression frameworks to retain experienced professionals and encourage long-term employee development.

7.5 Company Size Analysis

Objective

To determine whether organizational size influences employee compensation.

Visualization

Bar Chart – Average Salary by Company Size

(Insert chart.)

Interpretation

Large organizations generally provide higher average salaries than medium and small enterprises. This trend reflects greater financial capacity, larger AI investments, and increased demand for specialized technical expertise.

Business Insight

Professionals seeking higher earning potential may benefit from targeting larger organizations, while smaller companies should emphasize alternative benefits such as flexibility, innovation opportunities, and accelerated career growth.

7.6 Industry Analysis

Objective

To compare salary levels across industries.

Visualization

Bar Chart – Average Salary by Industry

(Insert chart.)

Interpretation

Industries with greater investment in AI technologies demonstrate significantly higher compensation than traditional sectors. Technology-driven organizations continue to lead the market in attracting highly skilled professionals.

Business Insight

Industry specialization has become an increasingly important factor in salary determination. Organizations operating in emerging AI sectors should maintain competitive compensation structures to remain attractive employers.

7.7 Geographic Salary Analysis

Objective

To investigate salary differences across countries and regions.

Visualization

Map / Bar Chart – Average Salary by Country

(Insert visualization.)

Interpretation

The analysis reveals considerable geographic variation in compensation. Countries with mature AI ecosystems generally offer higher salaries than developing markets.

Business Insight

International organizations should adopt region-specific salary benchmarking rather than applying standardized global compensation policies.

7.8 Remote Work Analysis

Objective

To examine the distribution of remote, hybrid, and on-site employment.

Visualization

Pie Chart – Remote Work Distribution

(Insert chart.)

Interpretation

The dataset indicates that remote employment has become an important component of the AI workforce. Hybrid work arrangements also represent a substantial proportion of available positions.

Business Insight

Flexible work policies have evolved into a competitive advantage for attracting and retaining AI professionals.

7.9 Certification Analysis

Objective

To determine whether professional certifications influence salary levels.

Visualization

Bar Chart – Average Salary by Certification Level

(Insert chart.)

Interpretation

Employees holding professional certifications generally receive higher average salaries than those without certifications.

Business Insight

Continuous professional learning enhances employability and contributes positively to long-term earning potential.

7.10 AI Tool Usage Analysis

Objective

To analyze the relationship between AI tool usage frequency and salary.

Visualization

Bar Chart – Average Salary by AI Tool Usage

(Insert chart.)

Interpretation

Professionals who regularly utilize AI tools tend to earn higher salaries than those with limited exposure to AI technologies.

Business Insight

Practical AI proficiency has become an increasingly valuable skill in the modern labor market and directly contributes to enhanced compensation opportunities.

7.11 Job Satisfaction Analysis

Objective

To examine employee satisfaction across the workforce.

Visualization

Histogram / Box Plot – Job Satisfaction

(Insert chart.)

Interpretation

Employee satisfaction scores indicate moderate overall satisfaction, although variations exist across different experience levels and industries.

Business Insight

Organizations should regularly monitor employee satisfaction and engagement to improve retention and workplace productivity.

7.12 Correlation Analysis

Objective

To identify relationships between numerical variables.

Visualization

Correlation Heatmap

(Insert heatmap.)

Interpretation

The correlation analysis highlights meaningful relationships between salary, experience, bonus percentage, AI tool usage, and job satisfaction.

Business Insight

Understanding these relationships enables organizations to identify the factors most strongly associated with employee compensation and performance.

7.13 Outlier Analysis

Objective

To detect unusually high or low salary observations.

Visualization

Box Plot – Salary Distribution

(Insert chart.)

Interpretation

Several high-income observations were identified, representing executive leadership and highly specialized AI positions. These outliers were retained because they reflect legitimate market conditions rather than data quality issues.

Business Insight

Outlier analysis provides a more realistic understanding of executive compensation and prevents misleading statistical conclusions.

7.14 Key Findings from Exploratory Data Analysis

The exploratory analysis revealed several important insights:

- Salary increases consistently with professional experience.
 - Large organizations generally provide higher compensation than smaller firms.
 - Technology-intensive industries offer the highest average salaries.
 - Geographic location significantly influences earning potential.
 - AI tool proficiency is associated with higher compensation.
 - Professional certifications positively impact salary levels.
 - Remote work has become a standard employment model within the AI sector.
 - Executive positions represent a small proportion of the workforce but account for the highest salaries.
 - Salary distribution is positively skewed due to high-income leadership roles.
 - Workforce analytics can support strategic hiring, compensation planning, and talent retention initiatives.
-

7.15 Chapter Summary

The Exploratory Data Analysis phase transformed raw employment records into actionable business intelligence by identifying compensation patterns, workforce characteristics, and organizational trends. Through statistical visualization and descriptive analysis, the project

demonstrated how experience, industry, company size, geographic location, and AI specialization collectively shape salary outcomes. These findings establish a strong analytical foundation for the SQL-based business analysis and executive dashboard presented in the following chapters.

CHAPTER 5

SQL Business Analysis

8. SQL Business Analysis

8.1 Introduction

Structured Query Language (SQL) is one of the most essential technologies in modern data analytics, enabling analysts to efficiently retrieve, transform, and analyze large volumes of structured data. In this project, MySQL was used to perform business-oriented analysis on the cleaned AI & Data Science Job Salaries 2026 dataset.

The objective of this phase was not only to execute SQL queries but also to answer real-world business questions related to compensation trends, workforce characteristics, organizational structures, and geographic salary variations. By leveraging SQL's analytical capabilities, the project demonstrates practical database skills that support data-driven decision-making.

8.2 Database Design

The cleaned dataset was imported into MySQL as a structured relational table. Appropriate data types were assigned to each column to ensure efficient storage, query performance, and analytical accuracy.

The database design focused on maintaining data integrity while supporting complex analytical queries involving aggregation, filtering, grouping, sorting, and ranking.

8.3 Business Questions

The SQL analysis was designed to answer the following strategic business questions:

- What is the average salary across all AI and Data Science professionals?
- Which job titles receive the highest average salaries?
- How does compensation vary by experience level?
- Which industries offer the most competitive salaries?
- Does company size influence employee earnings?
- Which countries provide the highest-paying opportunities?
- How does remote work impact compensation?
- Which programming languages are associated with higher salaries?
- Do certifications improve employee earnings?
- Which AI tools are most commonly used by high-income professionals?

These questions simulate the types of analytical requests commonly received by HR managers, business leaders, and executive stakeholders.

8.4 Data Retrieval

The initial SQL queries focused on understanding the overall structure of the dataset.

Typical operations included:

- Viewing sample records
- Counting total observations
- Identifying unique values
- Exploring available job roles
- Examining industry distribution

These exploratory queries provided a foundation for advanced business analysis.

8.5 Aggregate Analysis

Aggregate functions were used to summarize key business metrics.

The following calculations were performed:

- Average Salary
- Maximum Salary
- Minimum Salary
- Average Bonus Percentage
- Average Years of Experience
- Average Job Satisfaction
- Total Number of Employees

These metrics provide executive-level indicators of workforce compensation and organizational characteristics.

8.6 Salary Analysis by Experience Level

The analysis examined the relationship between professional experience and employee compensation.

The SQL results demonstrated a clear positive relationship between years of experience and salary, with Executive and Lead-level professionals consistently earning the highest compensation.

Business Insight

Organizations seeking to retain experienced AI professionals should establish transparent career progression frameworks and leadership development programs.

8.7 Salary Analysis by Company Size

This analysis compared compensation across small, medium, and large organizations.

The findings revealed that large organizations generally offer higher salaries due to greater financial resources, larger technology investments, and increased demand for specialized expertise.

Business Insight

Company size remains an important factor in salary benchmarking and workforce planning.

8.8 Industry Salary Analysis

Industry-level analysis identified significant differences in compensation across business sectors.

Technology-focused industries consistently demonstrated higher average salaries compared to traditional sectors.

Business Insight

Organizations operating in AI-intensive industries must maintain competitive compensation packages to attract and retain highly skilled professionals.

8.9 Geographic Salary Analysis

The SQL analysis compared salary levels across countries and regions.

The results highlighted substantial geographic variation in compensation, reflecting differences in economic development, labor market demand, and organizational investment in AI technologies.

Business Insight

International organizations should adopt region-specific salary benchmarking strategies rather than applying standardized global compensation policies.

8.10 Remote Work Analysis

Remote work has become an important characteristic of the modern AI workforce.

The SQL analysis examined salary differences among Remote, Hybrid, and On-site employees.

The findings indicate that remote professionals remain highly competitive in terms of compensation, demonstrating that organizations increasingly value flexibility without significantly reducing salary levels.

Business Insight

Flexible work arrangements can improve talent acquisition while maintaining competitive compensation structures.

8.11 Programming Language Analysis

Programming languages play a critical role in determining professional specialization.

The SQL analysis compared salary levels associated with different programming languages.

Professionals working primarily with Python and other advanced analytical technologies generally achieved higher average compensation.

Business Insight

Technical skill specialization continues to influence earning potential within the AI labor market.

8.12 Certification Analysis

Professional certifications were analyzed to evaluate their relationship with employee earnings.

The results indicate that certified professionals generally receive higher salaries than employees without recognized certifications.

Business Insight

Continuous professional development contributes positively to long-term career growth and compensation.

8.13 Advanced SQL Analysis

To strengthen analytical capabilities, several advanced SQL techniques were applied throughout the project.

These included:

- Common Table Expressions (CTEs)
- Window Functions
- Ranking Functions
- Dense Ranking
- Aggregate Functions
- GROUP BY Analysis
- ORDER BY Optimization

- HAVING Clause
- CASE Statements
- Subqueries

These techniques enabled the extraction of deeper business insights beyond basic descriptive analysis.

8.14 Executive Insights

The SQL analysis produced several important findings:

- Executive-level professionals receive substantially higher salaries than entry-level employees.
 - Large organizations consistently provide stronger compensation packages.
 - AI-focused industries offer the highest average salaries.
 - Geographic location significantly influences earning potential.
 - Professional certifications contribute positively to employee earnings.
 - Remote work has become a competitive employment model.
 - Technical specialization remains one of the strongest predictors of compensation.
 - Business intelligence techniques can effectively support workforce planning and salary benchmarking.
-

8.15 Chapter Summary

The SQL Business Analysis phase demonstrates how structured query language can transform raw employment records into actionable business intelligence. Through aggregation, filtering, ranking, and advanced analytical queries, the project identified the primary factors influencing compensation within the global AI and Data Science workforce. These findings complement the Python-based exploratory analysis and provide a strong foundation for executive-level reporting through Power BI dashboards.

CHAPTER 6

Power BI Dashboard Analysis

9. Power BI Dashboard Analysis

9.1 Introduction

Data visualization plays a crucial role in transforming complex datasets into actionable business intelligence. While statistical analysis provides numerical evidence, interactive dashboards enable executives and business leaders to monitor key performance indicators, identify emerging trends, and make informed decisions quickly.

To support strategic workforce analysis, an interactive Power BI dashboard was developed using the cleaned AI & Data Science Job Salaries 2026 dataset. The dashboard integrates multiple dimensions—including compensation, experience, company characteristics, geographic distribution, and remote work—into a unified decision-support system.

Designed with an executive audience in mind, the dashboard emphasizes simplicity, interactivity, and business storytelling. Dynamic filters allow users to explore salary trends across different industries, experience levels, company sizes, and geographic regions, making the dashboard suitable for HR professionals, recruiters, hiring managers, and organizational leaders.

9.2 Dashboard Objectives

The dashboard was designed to achieve the following objectives:

- Provide a consolidated overview of the AI and Data Science job market.
 - Monitor key workforce and compensation metrics.
 - Compare salary trends across industries, job roles, and company sizes.
 - Identify regional differences in compensation.
 - Analyze the impact of experience and remote work on salaries.
 - Support evidence-based hiring and compensation decisions.
 - Enable interactive exploration through dynamic filters.
-

9.3 Dashboard Layout

The dashboard follows a structured executive reporting layout divided into four analytical sections:

Executive KPI Panel

The top section presents high-level performance indicators, allowing users to quickly understand the overall characteristics of the dataset.

Workforce & Compensation Analysis

The central section focuses on salary comparisons across experience levels, company sizes, industries, and job roles.

Geographic Intelligence

An interactive map visual highlights the distribution of salaries across different company locations.

Interactive Filtering

Slicers enable users to dynamically filter results by:

- Industry
- Experience Level
- Company Size
- Company Location

This interactive functionality supports customized analysis without requiring technical expertise.

9.4 Executive KPI Analysis

The dashboard includes seven executive-level KPI cards that summarize the overall workforce profile.

KPI	Value	Business Interpretation
Total Jobs	5,000	Represents a robust dataset for market analysis.
Average Salary	\$98.61K	Serves as the benchmark compensation level across the dataset.
Average Experience	8.36 Years	Indicates a workforce dominated by mid-level and senior professionals.
Average Bonus	10.80%	Highlights the importance of performance-based incentives.

Average Satisfaction	6.69 / 10	Suggests moderate employee satisfaction with opportunities for improvement.
Maximum Salary	\$372.35K	Reflects compensation for highly specialized or executive AI roles.
Minimum Salary	\$12K	Represents entry-level or lower-cost regional positions.

Executive Observation

The significant gap between the average salary and maximum salary demonstrates a highly segmented compensation structure, where advanced expertise and leadership positions command substantial salary premiums.

9.5 Salary by Experience Level

Objective

To evaluate how professional experience influences annual compensation.

Visualization

(Insert Power BI Bar Chart: Average Salary by Experience Level)

Analysis

The visualization demonstrates a consistent increase in salary with higher experience levels. Executive professionals receive the highest compensation, followed by Lead and Senior employees, while Entry-Level positions occupy the lowest salary range.

Business Insight

Career progression remains one of the strongest predictors of compensation. Organizations should invest in employee development programs and leadership pipelines to improve retention and internal promotion.

9.6 Salary by Company Size

Objective

To compare employee compensation across different organizational sizes.

Visualization

(Insert Company Size Bar Chart)

Analysis

Large enterprises consistently offer the highest average salaries, reflecting greater investment capacity and demand for specialized AI talent. Medium-sized companies remain competitive, whereas smaller organizations generally provide lower compensation.

Business Insight

Company size should be considered when benchmarking salaries and developing recruitment strategies.

9.7 Geographic Salary Distribution

Objective

To visualize regional salary differences across company locations.

Visualization

(Insert Map Visual)

Analysis

The geographic analysis highlights substantial variation in compensation across regions. Countries with established technology ecosystems demonstrate higher salary levels compared to emerging markets.

Business Insight

Organizations expanding internationally should adopt localized compensation strategies that reflect regional labor market conditions.

9.8 Top 10 Highest Paying Job Titles

Objective

To identify the highest-paying positions within the AI and Data Science sector.

Visualization

(Insert Horizontal Bar Chart)

Analysis

Leadership-oriented and highly specialized AI roles dominate the highest-paying positions, emphasizing the market demand for advanced technical expertise and strategic leadership.

Business Insight

Professionals seeking higher earning potential should pursue specialized AI domains and leadership responsibilities.

9.9 Industry Salary Analysis

Objective

To compare compensation across different industries.

Visualization

(Insert Industry Bar Chart)

Analysis

Technology-intensive industries consistently outperform traditional sectors in salary levels due to greater investment in digital transformation and artificial intelligence.

Business Insight

Industry selection significantly influences long-term career growth and compensation.

9.10 Remote Work Analysis

Objective

To examine the distribution and compensation of Remote, Hybrid, and On-site roles.

Visualization

(Insert Donut Chart or Pie Chart)

Analysis

Remote employment represents a significant portion of the AI workforce. Hybrid work arrangements also remain popular, reflecting the evolving nature of modern workplaces.

Business Insight

Flexible work policies have become a competitive advantage in attracting and retaining highly skilled AI professionals.

9.11 Interactive Dashboard Features

The dashboard incorporates several interactive elements to enhance user experience:

- Dynamic slicers for multi-dimensional filtering.
- Cross-filtering between visuals.
- Responsive KPI cards.
- Interactive map visualization.
- Drill-down capability for detailed exploration.
- Executive-friendly layout for rapid decision-making.

These features allow stakeholders to perform customized analysis without requiring technical expertise.

9.12 Dashboard Strengths

The dashboard demonstrates several strengths aligned with Business Intelligence best practices:

- Clear and intuitive layout.
 - Consistent visual design.
 - Executive-focused KPI presentation.
 - Interactive exploration capabilities.
 - Effective use of geographic visualization.
 - Strong business storytelling.
 - Actionable analytical insights.
-

9.13 Recommended Enhancements

To further elevate the dashboard to enterprise standards, the following improvements are recommended:

- Add dynamic report titles that update based on selected filters.
 - Include median salary and salary percentile KPIs (P25, P50, P75, P90).
 - Develop drill-through pages for country and industry analysis.
 - Create tooltip pages with additional contextual information.
 - Incorporate trend indicators and conditional formatting.
 - Add bookmarks to enable executive presentation mode.
 - Optimize color consistency and spacing for enhanced readability.
-

9.14 Business Value

The Power BI dashboard provides substantial value for multiple stakeholders:

Human Resource Teams

- Salary benchmarking
- Workforce planning
- Recruitment optimization

Business Executives

- Strategic decision-making
- Compensation planning
- Talent management

Job Seekers

- Career planning
- Salary expectations
- Industry comparisons

Data Analysts

- Interactive business intelligence reporting
 - Executive dashboard design
 - Decision-support analytics
-

9.15 Chapter Summary

The Power BI dashboard successfully converts complex workforce data into a comprehensive executive reporting solution. By combining interactive visualizations, key performance indicators, and dynamic filtering capabilities, the dashboard enables stakeholders to explore salary trends, compare workforce characteristics, and support strategic business decisions. This chapter demonstrates proficiency in dashboard design, data modeling, DAX-driven reporting, and business storytelling—core competencies expected of modern Data Analysts and Business Intelligence professionals.

CHAPTER 7

Business Insights & Strategic Recommendations

10. Business Insights

10.1 Introduction

The objective of data analytics extends beyond creating visualizations or calculating summary statistics. The true value of analytics lies in transforming raw data into meaningful business insights that support strategic decision-making. Based on the exploratory data analysis, SQL business analysis, and Power BI dashboard, several significant patterns were identified within the AI and Data Science job market.

The following insights highlight the major factors influencing compensation, workforce structure, organizational behavior, and employment trends.

Insight 1: Experience is the Strongest Salary Driver

The analysis demonstrates a strong positive relationship between professional experience and annual salary. Employees at Senior, Lead, and Executive levels consistently earn substantially higher salaries than Entry-Level professionals.

As organizations increasingly rely on experienced AI specialists to lead innovation initiatives, compensation rises significantly with career progression.

Business Impact

Organizations should establish structured career development programs and leadership pipelines to retain experienced professionals and reduce turnover among high-value employees.

Insight 2: Large Organizations Offer Higher Compensation

Employees working in large organizations generally receive higher salaries compared to those employed by medium and small enterprises.

This trend reflects greater investment capacity, larger technology budgets, and increased demand for specialized AI talent.

Business Impact

Small and medium-sized organizations should compete through flexible work arrangements, career development opportunities, and performance-based incentives rather than salary alone.

Insight 3: AI-Driven Industries Lead the Salary Market

Industries with extensive investment in Artificial Intelligence consistently provide higher compensation than traditional sectors.

Organizations operating in technology, software, fintech, healthcare AI, and cloud computing demonstrate stronger demand for advanced analytical skills.

Business Impact

Professionals seeking long-term salary growth should prioritize industries undergoing rapid digital transformation.

Insight 4: Geographic Location Significantly Influences Compensation

Salary levels vary considerably across countries and regions.

Technology hubs generally provide significantly higher compensation due to stronger economic conditions, greater AI investment, and increased competition for skilled professionals.

Business Impact

Organizations operating globally should implement region-specific salary benchmarking instead of applying standardized global compensation structures.

Insight 5: Remote Work Has Become a Competitive Advantage

The analysis indicates that remote employment represents a substantial proportion of the AI workforce.

Contrary to traditional assumptions, remote professionals remain highly competitive in terms of compensation.

Business Impact

Organizations maintaining flexible work policies are better positioned to attract global talent while reducing recruitment barriers.

Insight 6: Professional Certifications Increase Career Value

Employees holding recognized certifications generally demonstrate stronger earning potential than professionals without certifications.

Continuous learning has become an important differentiator within the AI labor market.

Business Impact

Organizations should encourage certification programs to strengthen workforce capabilities while employees should invest in continuous professional development.

Insight 7: AI Tool Proficiency Enhances Compensation

Professionals who frequently utilize modern AI tools tend to receive higher salaries than those with limited exposure.

Technical specialization continues to influence long-term career success.

Business Impact

Organizations should invest in AI training initiatives to improve employee productivity and maintain competitive advantages.

Insight 8: Salary Distribution is Highly Uneven

The salary distribution exhibits significant positive skewness, indicating that while most professionals receive moderate salaries, a relatively small group of highly specialized experts earn exceptionally high compensation.

Business Impact

HR departments should utilize salary percentile analysis rather than relying solely on average salaries when designing compensation policies.

Insight 9: Employee Satisfaction Does Not Depend Solely on Salary

The analysis suggests that although compensation is an important factor, employee satisfaction is influenced by multiple variables including work flexibility, career development opportunities, organizational culture, and work-life balance.

Business Impact

Organizations should adopt holistic employee engagement strategies rather than focusing exclusively on financial rewards.

Insight 10: Data-Driven Compensation Planning Supports Better Hiring Decisions

The integration of Python, SQL, and Power BI demonstrates how analytical tools can improve salary benchmarking and workforce planning.

Organizations using business intelligence techniques are better equipped to make evidence-based hiring and compensation decisions.

Business Impact

Data-driven HR strategies reduce uncertainty and improve workforce planning efficiency.

11. Strategic Recommendations

11.1 Introduction

Based on the findings obtained throughout this project, the following strategic recommendations are proposed to improve talent acquisition, employee retention, compensation planning, and workforce development within AI-driven organizations.

Recommendation 1: Develop Competitive Salary Benchmarking Frameworks

Organizations should continuously monitor market salary trends and benchmark compensation against industry standards to remain competitive in attracting high-quality AI professionals.

Expected Benefit

- Improved recruitment success
 - Reduced employee turnover
 - Enhanced employer branding
-

Recommendation 2: Invest in Continuous Learning

Companies should encourage employees to pursue professional certifications, AI training programs, and technical skill development.

Expected Benefit

- Higher workforce productivity
 - Increased innovation
 - Improved employee retention
-

Recommendation 3: Strengthen Leadership Development Programs

Experienced AI professionals represent valuable organizational assets.

Companies should establish structured leadership development initiatives that prepare senior employees for strategic roles.

Expected Benefit

- Internal talent growth
 - Reduced external hiring costs
 - Improved succession planning
-

Recommendation 4: Expand Flexible Work Policies

Remote and hybrid work arrangements have become important competitive advantages.

Organizations should continue investing in digital collaboration technologies while maintaining flexible employment models.

Expected Benefit

- Access to global talent
 - Higher employee satisfaction
 - Improved work-life balance
-

Recommendation 5: Utilize Business Intelligence for HR Decisions

HR departments should integrate analytics platforms such as Power BI into workforce planning activities.

Interactive dashboards provide real-time insights that support evidence-based compensation and hiring decisions.

Expected Benefit

- Faster decision-making
 - Improved workforce planning
 - Better resource allocation
-

Recommendation 6: Focus on Skill-Based Hiring

Rather than emphasizing academic qualifications alone, organizations should prioritize practical AI skills, project experience, and technical expertise.

Expected Benefit

- Improved hiring quality
 - Higher employee performance
 - Stronger organizational competitiveness
-

Recommendation 7: Encourage Innovation Through AI Adoption

Organizations should increase investment in AI technologies while encouraging employees to utilize modern AI tools within daily operations.

Expected Benefit

- Greater operational efficiency
 - Improved decision-making
 - Increased business competitiveness
-

12. Business Impact

Organizational Benefits

This project demonstrates how data analytics can support business strategy in the following areas:

Human Resources

- Salary benchmarking
- Workforce planning
- Recruitment optimization
- Employee retention strategies

Executive Management

- Strategic compensation planning
- Resource allocation
- Talent acquisition decisions
- Market competitiveness analysis

Business Intelligence Teams

- Executive reporting
- KPI monitoring
- Interactive dashboards
- Decision-support analytics

Employees and Job Seekers

- Career planning
- Salary negotiation
- Skill development
- Industry comparison

Overall Business Value

The project successfully transforms raw employment data into actionable business intelligence that enables organizations to make informed, evidence-based decisions regarding compensation, workforce planning, and talent management. By integrating Python, SQL, and Power BI, the analysis demonstrates an end-to-end analytics workflow capable of supporting executive decision-making and strategic business planning.

CHAPTER 8

Conclusion, Future Scope, References & About the Author

13. Conclusion

13.1 Project Summary

The rapid expansion of Artificial Intelligence and Data Science has significantly reshaped the global employment landscape, creating a growing demand for highly skilled professionals. Understanding salary trends, workforce characteristics, and compensation drivers has become increasingly important for organizations seeking to attract and retain top talent.

This project presented a comprehensive end-to-end Business Intelligence analysis of the **AI & Data Science Job Salaries 2026** dataset using Python, MySQL, and Microsoft Power BI. The analytical process included data preparation, feature engineering, exploratory data analysis, SQL-based business intelligence, and executive dashboard development.

Throughout the project, multiple analytical techniques were applied to transform raw employment records into meaningful business insights. The findings demonstrated that

experience level, organizational size, industry specialization, geographic location, professional certifications, and AI tool proficiency are among the most influential factors affecting employee compensation.

The integration of Python enabled efficient data cleaning, preprocessing, statistical analysis, and visualization. SQL facilitated business-oriented querying and workforce analysis, while Power BI transformed analytical findings into an interactive executive dashboard that supports strategic decision-making.

Rather than focusing solely on technical implementation, this project emphasized business storytelling by converting analytical results into actionable recommendations for HR professionals, recruiters, executives, and business leaders.

Overall, the project successfully demonstrates the complete data analytics lifecycle and reflects practical proficiency in Business Intelligence, Data Analysis, SQL, and dashboard development. It also highlights the ability to bridge the gap between technical analysis and strategic business decision-making.

13.2 Key Achievements

During this project, the following objectives were successfully accomplished:

- Conducted comprehensive data cleaning and validation.
 - Performed feature engineering to improve analytical depth.
 - Completed detailed exploratory data analysis using Python.
 - Executed business-oriented SQL queries to answer strategic questions.
 - Developed an interactive executive dashboard in Microsoft Power BI.
 - Identified major salary drivers within the AI job market.
 - Generated actionable business insights and strategic recommendations.
 - Demonstrated an end-to-end analytics workflow aligned with industry best practices.
-

13.3 Final Reflection

This project demonstrates that effective data analytics extends beyond technical programming. The true value of analytics lies in transforming complex datasets into clear, actionable, and evidence-based business insights that support strategic organizational decisions.

By combining Python, SQL, and Power BI within a structured analytical framework, the project illustrates how modern Business Intelligence solutions can improve workforce planning, compensation benchmarking, talent acquisition, and organizational decision-making.

14. Future Scope

Although this project provides comprehensive descriptive and business intelligence analysis, several opportunities exist for future enhancement.

14.1 Machine Learning Salary Prediction

Future work may include developing predictive models using regression algorithms to estimate salaries based on experience, education, certifications, industry, company size, and geographic location.

Potential algorithms include:

- Linear Regression
 - Random Forest Regression
 - XGBoost
 - CatBoost
-

14.2 Interactive Web Application

The project can be expanded into a web-based analytics platform using:

- Streamlit
- Flask
- Django

This would enable recruiters and job seekers to perform interactive salary analysis through a browser interface.

14.3 Cloud Deployment

Future versions may be deployed using cloud platforms such as:

- Microsoft Azure
- AWS
- Google Cloud Platform

Cloud deployment would improve scalability, accessibility, and collaboration.

14.4 Real-Time Data Integration

Rather than relying on historical datasets, future implementations may integrate APIs from job portals to collect real-time salary information.

Possible data sources include:

- LinkedIn Jobs
 - Indeed
 - Glassdoor
 - RemoteOK
 - Wellfound
-

14.5 Advanced HR Analytics

Additional analytical modules may include:

- Employee Attrition Prediction
 - Workforce Planning
 - Hiring Demand Forecasting
 - Skill Gap Analysis
 - Recruitment Performance Analysis
-

14.6 Generative AI Integration

Modern AI assistants can be integrated into the dashboard to automatically generate:

- Executive summaries
 - Salary insights
 - Hiring recommendations
 - Workforce reports
 - Natural language explanations
-

14.7 Interactive Power BI Enhancements

Future dashboard improvements may include:

- Drill-through reports
- Dynamic tooltips
- Bookmarks
- AI-powered visuals
- Forecasting visuals
- What-if analysis

- Decomposition Tree
 - Key Influencers Visual
-

15. References

The following resources were used throughout the development of this project.

Dataset

- Kaggle. (2026). *AI & Data Science Job Salaries 2026 Dataset*.

Programming Libraries

- Python Software Foundation. *Python Documentation*.
- Pandas Development Team. *Pandas Documentation*.
- NumPy Developers. *NumPy Documentation*.
- Matplotlib Development Team. *Matplotlib Documentation*.
- Seaborn Development Team. *Seaborn Documentation*.

Database

- Oracle Corporation. *MySQL Reference Manual*.

Business Intelligence

- Microsoft. *Power BI Documentation*.

Development Environment

- Microsoft. *Visual Studio Code Documentation*.

Version Control

- Git Documentation
 - GitHub Documentation
-

16. About the Author

Rana Muhammad Daniyal Farooq

Aspiring Data Analyst | Business Intelligence Enthusiast | Python • SQL • Power BI

Rana Muhammad Daniyal Farooq is an aspiring Data Analyst with a strong interest in Business Intelligence, Data Analytics, and Artificial Intelligence. He specializes in transforming raw data into actionable business insights through Python, SQL, and Microsoft Power BI.

His technical expertise includes data cleaning, exploratory data analysis, dashboard development, SQL-based business analysis, and data visualization. He is passionate about solving real-world business problems using analytical thinking and evidence-based decision-making.

This project demonstrates his ability to execute an end-to-end analytics workflow, from data preparation and feature engineering to executive dashboard development and strategic business reporting. It reflects his commitment to continuous learning and his aspiration to contribute to data-driven organizations.

Technical Skills

- Python
- SQL (MySQL)
- Power BI
- Microsoft Excel
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Git & GitHub

Career Objective

To build scalable, data-driven solutions that help organizations make informed business decisions while continuously expanding expertise in Data Analytics, Business Intelligence, Machine Learning, and Artificial Intelligence.